

1. ★ Seven Key Elements of a System

A system has seven key elements that work together to reach a goal.

Input is the data or resource entering the system.

Processor is the operational part that converts input into output.

Output is the final result produced.

Control is the decision-making part that manages and directs the system's activities.

Feedback is the information returned after output, used to improve future performance.

Environment includes external factors that affect the system but are not part of it.

Boundary defines what is inside and what is outside the system.

For example, in an online food delivery system, the customer's order is the input, the restaurant preparing food is the processor, delivering the pizza is the output, and the customer's rating is the feedback.

2. System Definition – Open and Closed System

A system is a set of components that interact together to achieve a common purpose, like a business, a language, or the human body. Systems can be open or closed.

An open system interacts with its environment, receiving inputs and delivering outputs to the outside world.

Most real-life systems, such as a university, hospital, or bank, are open systems because they constantly exchange information and resources with their surroundings.

A closed system does not interact with its environment at all; it is isolated from outside influences. A completely closed system is rare in real life, but a sealed thermos flask is often given as a hypothetical example.

3. Control Element of a System

The control element is the most essential part of any system because it keeps the system working correctly. It compares the actual performance of the system with a set standard, and if there is too much deviation, it makes adjustments. The control model has four parts: a standard to follow, performance measurement metrics, techniques to compare performance, and feedback. After comparison, the results are sent back to the system's input so corrections can be made. Without control, a system cannot stay balanced or reach its intended objective properly.

4. ★ Transaction Processing System (TPS)

A Transaction Processing System is a type of information system used to manage and process daily business transactions, collecting and storing both customer and business data. TPS has three main functions: system runtime functions that process transactions quickly while keeping data accurate and secure, system administration functions that handle configuration and monitoring, and application development functions that customize the system for specific business needs. A TPS has four components: input (orders, invoices, bills), processing (reading and updating data), output (receipts, reports), and storage (saving data in a database). For example, in a hospital TPS, patient details are the input, verifying and registering the patient is the processing, the registration slip and bill are the output, and all records are saved in the hospital database as storage.

5. Role of System Analysts

A system analyst can play different roles depending on the size of the organization. In **System Analysis**, the analyst collects facts and information through system studies to determine requirements. In **System Analysis and Design**, the analyst does the full system study along with developing design specifications, usually working on fewer projects. In **System Analysis, Design and Programming**, the analyst handles investigation, design, and also implements the system by programming it. Smaller companies usually give all these roles to one analyst, while larger companies hire separate specialists for each role.

6. Challenges for IS Developers

Developing an information system means channeling human knowledge and ideas to make a machine perform specific tasks efficiently. A good information system must function correctly, be easy to use, and help people work more efficiently. The biggest challenge for developers is making sure the technology does not become an obstacle, even for the most inexperienced or novice user, so that everyone can get the desired result without difficulty.

7. Benefits of Integrating Security into SDLC

Adding security into the System Development Life Cycle from the beginning gives several benefits. It allows early identification and fixing of security problems, which lowers overall costs. It creates awareness of engineering challenges caused by security controls, and allows reuse of shared security tools, reducing development cost. It supports better executive decision-making through proper risk management, and documents important security decisions for future reference. It also improves customer and organizational confidence in the system, and makes system integration easier since security is considered from the start rather than separately at different levels.

8. ★ Risk Management Process

Risk management is the systematic process of identifying, analyzing, evaluating, controlling, and continuously monitoring risks to reduce their impact on an organization or project. When applied to system development, it helps balance the need to protect information and assets against the cost of security controls throughout the SDLC. This process helps identify critical assets, operations, and vulnerabilities across the whole organization, allowing the organization to make informed decisions about where to invest in protection during system development.

9. ★ Development/Acquisition Phase (SDLC)

In this phase, the system is actually designed, purchased, programmed, or built. A key security activity here is conducting a risk assessment and using its results to add proper baseline security controls. The organization analyzes security requirements, performs functional and security testing, and prepares documents for certification. Security plans are also created, which describe the selected controls, the reasoning behind them, and how the system should be restricted in high-risk situations. Before moving to implementation, developmental testing ensures the technical and security features work as intended.

10. Operations/Maintenance Phase

In this phase, the system is already in place and running, while enhancements or modifications are developed and tested as needed. Hardware and software components may be added or replaced over time. The organization continuously monitors the system's performance to make sure it still meets the original user and security requirements. Configuration management activities are carried out to document any proposed or actual changes to the security plan, since systems constantly evolve with upgrades and environmental changes.

11. Disposal Phase

This is the final phase, where plans are made to discard system information, hardware, and software while transitioning to a new system. The information, hardware, and software can be moved to another system, archived, discarded, or destroyed. If this phase is done improperly, it can lead to unauthorized disclosure of sensitive data. Special care is given to properly preserving important data so it can be migrated or archived following the organization's records management policies, allowing possible future access if needed.

12. ★ Constraints of a System (asked in previous CT)

A system must follow three basic constraints to function properly. First, the system must have a defined structure and behavior designed to achieve a specific, predetermined objective. Second, there must be interconnectivity and interdependence among all the components of the system, meaning they rely on each other to work correctly. Third, the objectives of the whole organization must always be given higher priority than the objectives of any individual subsystem. Examples of such systems include a traffic management system, payroll system, and automatic library system.

13. ★ Properties of a System

A system has five main properties. **Organization** means the structure and order of components arranged to achieve set objectives. **Interaction** is how different components work together, like how a purchasing department interacts with the production department. **Interdependence** means components depend on one another, where the output of one subsystem becomes the input for another. **Integration** refers to how components are connected so they work together within the system despite performing unique functions. **Central Objective** means the system must have one clear main goal, though sometimes an organization states one objective but actually operates to achieve a different one.

14. Categories of Information

Information in a system can be divided into three categories based on management level. **Strategic Information** is needed by top management for long-range planning, such as revenue trends and investment decisions, and is supported by Decision Support Systems (DSS). **Managerial Information** is needed by middle management for short to intermediate planning, like sales analysis and cash flow projections, supported by Management Information Systems (MIS). **Operational Information** is needed by lower-level management for daily operations, such as attendance records and stock levels, supported by Data Processing Systems (DPS).

15. ★ Man-Made Information System (asked in previous CT)

A man-made information system is an interconnected set of information resources used to manage data for a particular organization under Direct Management Control (DMC). It includes hardware, software, communication, data, and applications needed to produce information according to organizational needs. There are three types. **Formal Information System** follows the official flow of information through memos and instructions from top management down to lower levels. **Informal Information System** is an employee-based system used to solve day-to-day work problems. **Computer-Based System** directly depends on computers to manage business applications, such as an automatic library system, railway reservation system, or banking system.